Customer Shopping Behaviour Analysis

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1. Project Overview

This project analyses the behaviour of customers using transactional data across 3,900 purchases across various product categories. The objective of this analysis was to discover insights regarding customer purchasing patterns, customer segments, product preferences, and subscription behaviour to inform strategic business decisions.

2. Dataset Summary

Rows: 3,900
Columns: 18
Key Features:

- Customer Demographics (Age, Gender, Location, Subscription Status)
- Purchase Details (Item Purchased, Category, Purchase Amount (USD), Size, Colour, Season, Payment Method)
- Shopping Behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Missing Data: 37 missing values in Review Rating column.

3. Exploratory Data Analysis and Cleaning using Python

The data was cleaned and initial exploration was conducted as follows:

Data Loading: Data was imported using the *pandas* library.
Initial Exploration: Examined a sample of the uncleaned dataset using df.head()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Pay M
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	V
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	I
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	I

Previous Purchases	Payment Method	Frequency of Purchases
14	Venmo	Fortnightly
2	Cash	Fortnightly
23	Credit Card	Weekly
49	PayPal	Weekly
31	PayPal	Annually

Utilised df.info() and df.describe() to conduct an initial exploration of the data structure and associated summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

Handling of Missing Data: Identified 37 null values in the *Review Rating* column and imputed the median review rating for each category.

Column Standardisation: Formatted each column name as snakecase. The column purchase_amount_(usd) was renamed to purchase_amount_usd.

Feature Engineering:

- Created a new column, *age_group* by binning customer ages.
- Created a new column, *purchase_frequency_days* to represent the *Frequency of Purchases* column numerically in days (i.e. Every 3 months as 90 days).

Data Consistency Check: Verified that *discount_applied* and *promo_code_used* columns were identical. Hence *promo_code_used* was removed as a redundant column.

Database Integration: Connected cleaned data in Python to a PostgreSQL database and loaded the dataset for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Structured analysis was conducted using PostgreSQL to answer key business questions:

1. **Revenue by Gender** - Performed a comparison of revenue generated by male vs. female customers.

gender text	revenue numeric
Female	75191
Male	157890

2. **High-spending Discount Users** - Identified customers who spent more than the average purchase ($59.76) amount despite using a discount.

	customer_id bigint	purchase_amount bigint	discount_applied text
1	2	64	Yes
2	3	73	Yes
3	4	90	Yes
4	7	85	Yes
5	9	97	Yes
6	12	68	Yes
7	13	72	Yes
Total rows: 839	Query complete 00:00:00.178		

3. **Top 5 Products by Rating** - Identified the top 5 products with the highest average review rating.

	item_purchased text	avg_review_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** - Compared average purchase amounts between Standard vs. Express shipping.

	shipping_type text	avg_purchase_amount numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribed vs. Non-Subscribers** - Compared the number of customers, the average purchase amount, and the total revenue between subscribers vs. non-subscribers.

	subscription_status text	count bigint	avg_purchase numeric	total_purcase numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

6. **Discount-Dependant Products** - Identified the top 5 products with the highest percentage of purchases with discounts applied.

	item_purchased text	percentage_discounted numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** - Classified customers as New, Returning, and Loyal depending on customer purchase history as follows:
   New - 1 previous purchase
   Returning - Between 2 and 10 (inclusive) previous purchases.
   Loyal - Greater than 10 previous purchases.

	customer_loyalty text	number_of_customers bigint
1	Loyal	3116
2	Returning	701
3	New	83

8. **Top 3 Products Per Category** - Identified the 3 most purchased products within each category.

	category text	item_purchased text	item_count bigint	category_item_ranking bigint
1	Accessori...	Jewelry	171	1
2	Accessori...	Sunglasses	161	2
3	Accessori...	Belt	161	3
4	Clothing	Blouse	171	1
5	Clothing	Pants	171	2
6	Clothing	Shirt	169	3
7	Footwear	Sandals	160	1
8	Footwear	Shoes	150	2
9	Footwear	Sneakers	145	3
10	Outerwear	Jacket	163	1
11	Outerwear	Coat	161	2

Total rows: 11 Query complete 00:00:00.123

9. **Repeat Buyers & Subscriptions** - Identified if customers with greater than 5 previous purchases are more likely to subscribe.

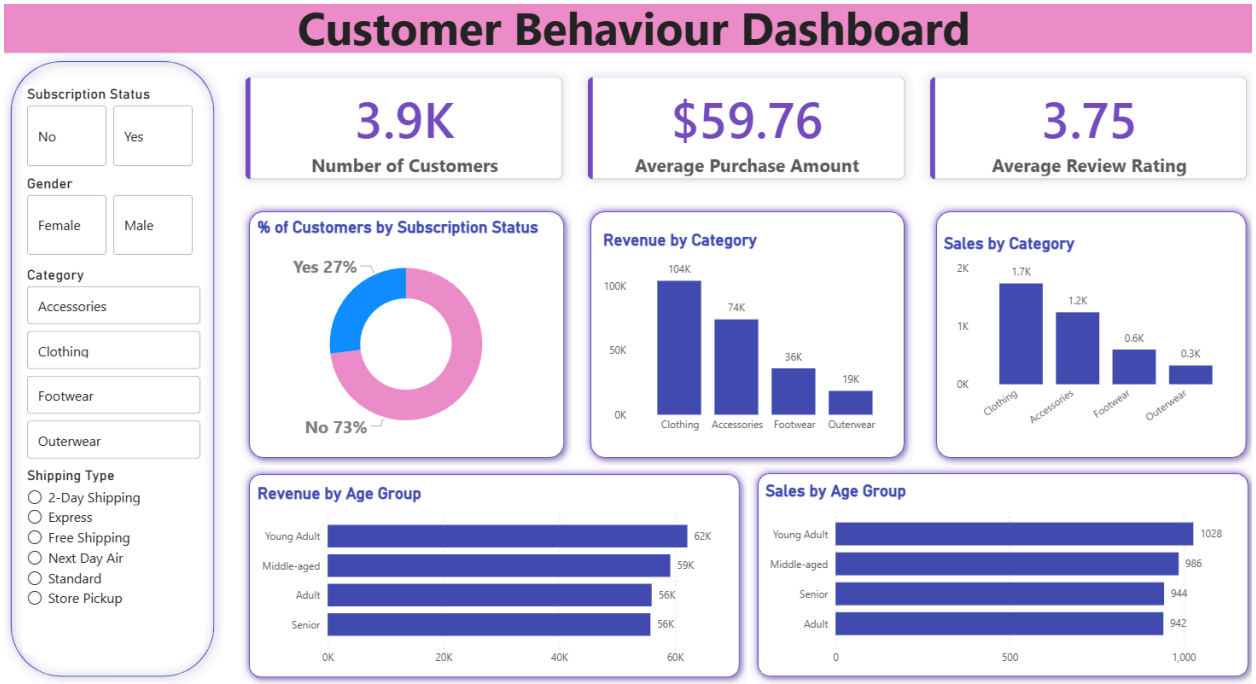
	subscription_status text	subscription_count bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** - Calculated the total revenue contributed by each age group.

	age_group text	revenue_contribution numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

A dashboard was produced in PowerBI to present insights in a visual format.



6. Business Recommendations

Boost Subscriptions - Promote exclusive benefits for subscribers to incentivise new subscriptions.

Customer Loyalty Programs - Reward repeat customers to incentivise them to subscribe.

Review Discount Policy - Balance increased revenues from discounts with margin control.

Product Positioning - Highlight top-rated, and best-selling products in marketing campaigns.

Targeted Marketing - Focus efforts on high-revenue age groups and express-shipping users.